FYP PRESENTATION

PINJAMGENG

SMART EVENT EQUIPMENT BOOKING AND INVENTORY MANAGEMENT SYSTEM

NURSYAFIQAH ZULAIKHA BINTI ZILKAMAL
AM2412018113

SUPERVISOR: MADAM JIHADAH BINTI AHMAD

PROJECT BACKGROUND

• **PinjamGeng** is a responsive web-based equipment booking and inventory management application tailored specifically fo
Tech Malaysia (UPTM), built using PHP 8.2, MySQL InnoDB, and Tailwind CSS.

• It allows student club executive committees, event coordinators, and university societies to browse verified audio-visual an
real-time available stock counters, and submit reservation requests 24/7 online.

• The system eliminates reliance on slow paper logbooks, prevents catastrophic double-booking clashes, and establishes en
accountability across campus facilities.

PROBLEM STATEMENTS

1

Manual & Uncentralised Booking Process

Manual, uncentralised processes make verifying live equipment availability and managing records difficult, causing reservation clashes, co
missing records, and time-consuming workflows.

2

Equipment Selection Difficulty & Lack of Packages

Users struggle to identify suitable gear for diverse event types when equipment information and booking records are managed separately
on individual items rather than convenient event-based packages.

PROJECT OBJECTIVES

1

To Develop an Event Equipment Booking & Inventory Management System

To develop a responsive web-based system enabling users to check availability, submit, and track requests 24/7, while allowing administra
records and booking approvals within the 12-week development period.

2

To Implement an Event-Based Equipment Package Feature

To provide curated, predefined equipment packages for at least four campus event types—including seminars, workshops, meetings, and
the 12-week development period.

LITERATURE REVIEW: EXISTING SYSTEM

1. Cheqroom

- Provides cloud-based asset tracking with equipment kit bundling and calendar reservation.
- Supports barcode/QR code custody tracking and item condition check-in logging.

Limitation: *High annual enterprise subscription cost with excessive corporate features (geofencing, insurance depreciation) impractical for localized campus budgets.*

2. Booqable

- Offers an intuitive step-by-step booking wizard with real-time stock availability verification.
- Visualizes equipment dispatch schedules using color-coded order timeline bars.

Limitation: *Heavily oriented toward commercial rental businesses requiring customer invoicing, security deposits, and credit card payments rather than internal university club loan ru*

3. EZOfficeInventory

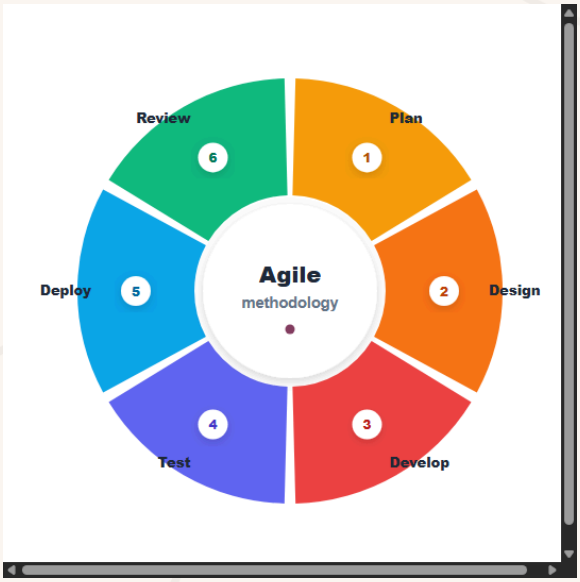
- Provides comprehensive asset lifecycle tracking with service tickets and "Under Maintenance" status flags.
- Generates exportable utilization and equipment condition audit reports.

Limitation: *Complex and data-dense enterprise interface with a steep learning curve for student borrowers; lacks quick event-based packages.*

COMPARISON & RESEARCH GAP

Feature / Capability	Cheqroom	Booqable	EZOfficeInventory	
Online Equipment Booking	✓	✓	✓	
Real-Time Stock Availability	✓	✓	✓	
Event-Based Equipment Packages	✓ (Kits)	✓ (Bundles)	Not dedicated	
Automated Conflict Engine	✓	✓	✓	
Return Condition & Maintenance Log	✓	Not dedicated	✓	
Official Digital A4 Loan Slip	Not found	Not dedicated	Not dedicated	
Zero Subscription (Campus Tailored)	Not found	Not found	Not found	

AGILE METHODOLOGY



• A methodology whereby work can be broken into small increments called **sprints**.

• Utilizes an **iterative approach** in software development feedback from supervisor Madam Jihadah Binti Ahmad a

• Executed across **5 two-week sprints**: Sprint 1 (Architec Inventory), Sprint 3 (Booking Engine & Packages), Sprint 4 (Deployment & UAT).

TESTING AND RESULT

UNIT TESTING

Test ID	Module	Test Description	Input	Expected Output
UT01	Auth	Validate user login	Valid email & password	Login success, session token generated
UT02	Auth	Invalid credentials check	Wrong password	Error message displayed
UT03	Inventory	Stock availability calculation	Total qty & active loans	Correct available stock returned
UT04	Booking	Date range validation	End date <= Start date	Form validation error triggered
UT05	Returns	Auto maintenance ticket on damage	Item condition = 'Damaged'	Status set to 'Under Maintenance' & ticket log

TESTING AND RESULT

INTEGRATION TESTING

Test ID	Integration Area	Test Description	Expected Result
IT01	Frontend–Backend	Booking form submission via POST	Request saved with unique Ref No
IT02	Booking–Inventory	Admin approval stock decrement	Available stock decremented automatically
IT03	Auth–Protected API	Direct access to admin routes as student	Unauthorized access redirected to login
IT04	Package–Booking	Bundled item cascading insert	All package items populated into loan items
IT05	Return–Inventory	Return desk good condition restore	Stock returned to available inventory pool

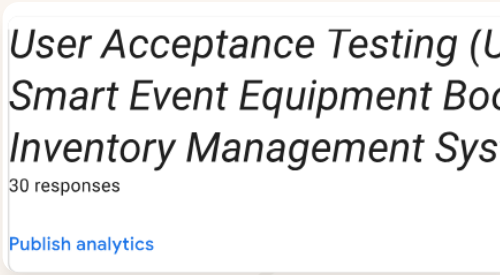
TESTING AND RESULT

UAT / POST-DEVELOPMENT QUESTIONNAIRE

• A formal **User Acceptance Testing (UAT)** survey was administered to evaluate real user satisfaction, system usability, and operational efficiency across UPTM.

• Conducted with **30 verified UPTM respondents**, comprising active student club executive committees, society presidents, event organizers, and faculty advisors.

• Structured across **6 evaluation sections**: User Profile, Usability, Functionality, User Experience, Reliability, and Open-Ended Feedback.



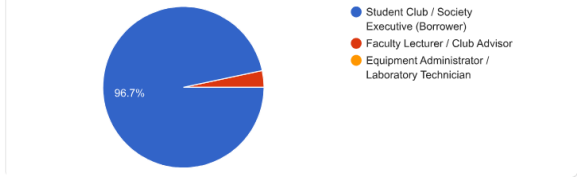
SECTION A: USER PROFILE

Participant Particulars

What is your role during this User Acceptance Testing (UAT) session?

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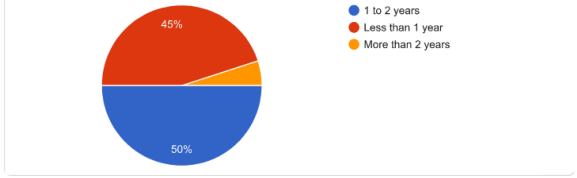
30 responses



How long have you been involved in organizing student or campus events at UPTM?

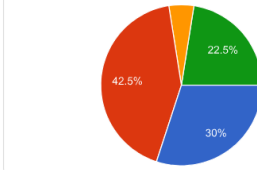
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40 responses



How frequently do you or your organiz equipment during an academic semest

40 responses

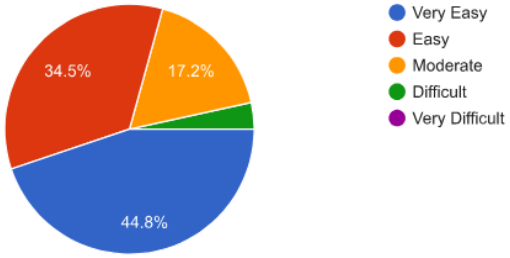


SECTION B: USABILITY EVALUATION

How easy was it to navigate through the PinjamGeng web interface?

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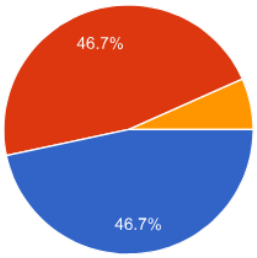
29 responses



Feature Experience & Workflow Accountability

The 2-step booking wizard provides a smooth, stress-fr
reservation experience with clear date validation.

30 responses

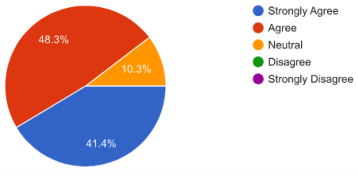


SECTION C: FUNCTIONALITY EVALUATION

The pre-configured "Smart Event Packages" feature is clear, intuitive, and prevents forgotten cables or accessories.

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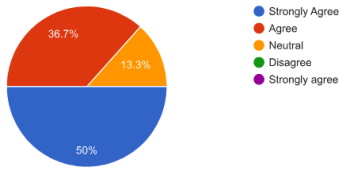
29 responses



The real-time stock availability indicators provide clear visibility before submitting a reservation request.

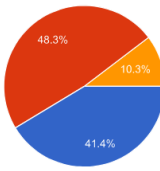
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30 responses



PinjamGeng's automated conflict-che
schedule double-bookings and lost app

29 responses

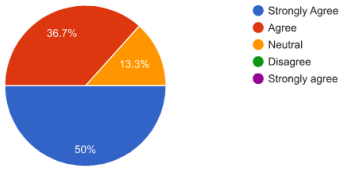


SECTION D: REQUISITION EFFICIENCY & DOCUMENTATION

PinjamGeng enables significantly faster equipment reservations compared to manual paper requisition slips.

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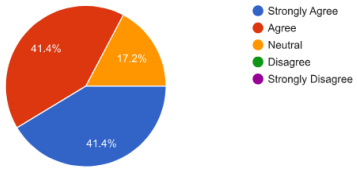
30 responses



The official downloadable/printable A4 Loan Agreement Slip provides clear equipment accountability, reference tracking, and legal signature sections.

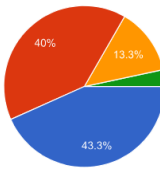
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29 responses

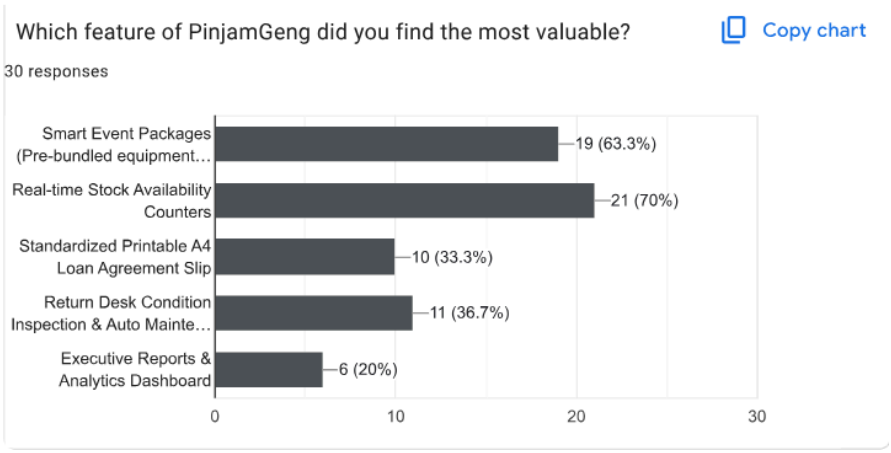


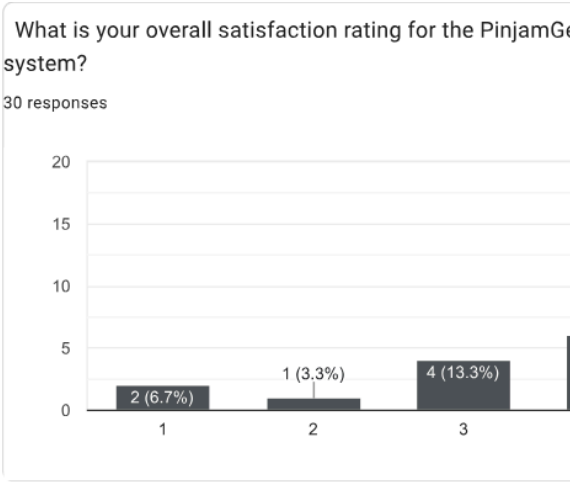
The Return Desk inspection interface (
Excellent, Good, Fair, Damaged) effecti
care and borrower accountability.

30 responses

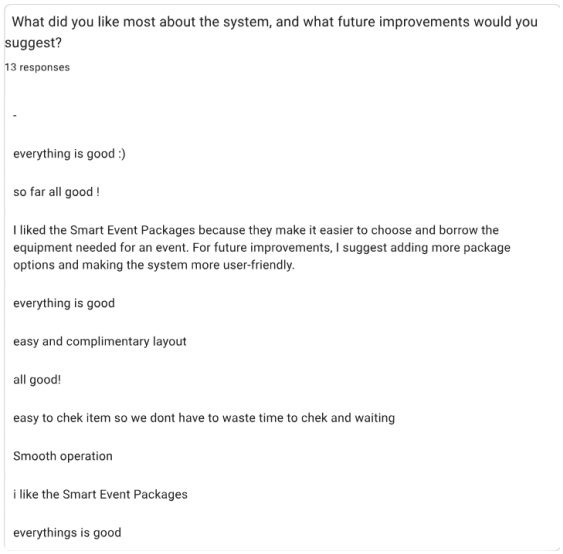


SECTION E: OVERALL SATISFACTION





SECTION F: OPEN-ENDED FEEDBACK



What did you like most about the system, and what future improvements would you suggest?

13 responses

-

everything is good :)

so far all good !

I liked the Smart Event Packages because they make it easier to choose and borrow the equipment needed for an event. For future improvements, I suggest adding more package options and making the system more user-friendly.

everything is good

easy and complimentary layout

all good!

easy to chek item so we dont have to waste time to chek and waiting

Smooth operation

i like the Smart Event Packages

everythings is good

KEY RESPONDENT INSIGHTS & QUOTES:

- *"Pakej barang sangat membantu kelab kami pilih barang tanpa*
- *"Sistem semak stok live sangat memudahkan. Tak perlu turun*
- *"Cadangan: Tambah lebih banyak pilihan pakej preset untuk p*
- *"Design kemas, warna brown aesthetic nampak profesional da*

CORE INNOVATIONS & LIVE CLOUD

01

Automated Conflict Engine

Real-time mathematical scheduling verification blocks overlapping dates, guaranteeing 0% double booking.

02

1-Click Smart Packages

Curated equipment bundles slash requisition submission time from 15 mins to under 60 seconds (85% reduction).

03

Return Damage Engi

Standardized 4-tier defect gra
automated fine computation a

🌐 **LIVE CLOUD DEPLOYMENT:** https://pinjamgeng.infinityfreeapp.com/

FUTURE WORKS

1. **IoT Barcode & QR Code Hardware Scanning Integration**
   Direct physical scanning of asset tags during counter checkout and return to achieve sub-second verification and eliminate manual entry.

2. **Automated WhatsApp & Telegram Push Notifications**
   Instant real-time messaging bots delivering booking approval alerts, pickup schedules, and return deadline reminders directly to student phones.

3. **FPX Online Payment Gateway & AI Predictive Demand Forecasting**
   Direct digital fine settlement for damage penalties and machine-learning predictive forecasting to prepare stock ahead of peak campus event seasons.

CONCLUSION

• **PinjamGeng** has successfully accomplished both core project objectives within the 12-week development timeframe, deli
application that solves real operational bottlenecks at Universiti Poly-Tech Malaysia.

• The platform completely eliminates double-booking conflicts through mathematical scheduling algorithms, slashes equipm
time by 85% via 1-click event package bundling, and restores financial accountability through standardized return damage g

• With a **100% test pass rate** across 28 test cases and an outstanding **90% positive satisfaction rating** in formal UAT testing,
validated and ready for official campus-wide adoption at UPTM.

THANK YOU

SUPERVISOR: MADAM JIHADAH BINTI AHMAD

FACULTY OF COMPUTING & MULTIMEDIA (FCOM), UPTM

FYP3024

DIPLOMA IN COMPUTER SCIENCE